Erdős Problems 249 and 257 in Lean 4:

formalised results, certificate reductions, and open questions

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Abstract

We formalise in Lean 4 several results around Erdős Problems 249 and 257. For every integer base $b \geq 2$, the development verifies the irrationality of the Erdős–Borwein series $\sum_{n\geq 1}(b^n - 1)^{-1}$, together with several structured infinite-support variants. For the totient constant $S = \sum_{n\geq 1} \varphi(n)/2^n$, whose irrationality remains open, we prove that any rational representation has denominator greater than $Q_0 \approx 7.96 \times 10^{34}$, and formalise an exact reduction of irrationality to an unbounded family of finite decidable certificates. We also describe the proof architecture and auxiliary rationality criteria based on binary carry systems. This work solves neither Erdős #249 nor the universal form of Erdős #257.

Keywords. irrationality; Erdős–Borwein constant; Euler totient; Lambert series; binary carry systems; Lean 4; Mathlib; formal verification.

MSC 2020. 11J72 (primary); 11A25, 68V20 (secondary).

1 Introduction

Two open questions provide the setting; their original formulations occur in the Erdős–Graham problem collection and Erdős's later survey [3, 4], while the numbering follows the maintained Erdős Problems catalogue [19]. Erdős #249 asks whether the totient constant

$$S = \sum_{n\geq 1} \frac{\varphi(n)}{2^n}$$

is irrational. Erdős #257 asks whether $\sum_{n\in A}(2^n - 1)^{-1}$ is irrational for every infinite $A \subseteq \mathbb{N}$. The second problem contains the classical full-support Erdős–Borwein constant, but the passage from structured supports to an arbitrary infinite support remains open.

The development has four principal outputs. First, it formalises the irrationality of $\sum_{n\geq 1}(b^n - 1)^{-1}$ for every integer $b \geq 2$, and of several structured infinite-support variants. Second, it places S on the same Mersenne–Lambert ladder and proves useful positive, signed, and probabilistic representations. Third, it gives the unconditional denominator bound $q > Q_0$ for any rational expression $S = p/q$. Fourth, it proves an exact finite-certificate reduction: irrationality follows if the certified non-integrality witnesses occur at unbounded parameters, and the existence of a witness at fixed parameters is equivalent to the corresponding tail difference being non-integral.

Scope. The irrationality of S and the universal infinite-support statement remain open. The denominator theorem and certificate reduction are genuine unconditional and conditional advances toward #249, respectively; neither is presented as a solution. The support theorems establish named families toward #257, not the universal quantifier.

The formalisation-specific contribution is not merely a list of declarations. It isolates a common architecture in which an infinite irrationality question is converted into finite arithmetic checked by the Lean kernel, while the remaining analytic uniformity statement is exposed rather than hidden inside computation. Exact links from the informal statements to the pinned source are kept as quiet hyperlinks; the complete declaration map is in Appendix B.

The prior-art boundary is theorem-specific. The full-support theorem is due to Erdős [1]; Borwein later proved a broader irrationality theorem that includes the relevant Mersenne–Lambert specialisation [5]. The pairwise-coprime support condition is also classical [2]. Periodic-coefficient Lambert series have a separate literature [11]. Kovač–Tao record the strict-tail geometry used by the achievement-set appendix [12], while Wang–Grau Ribas use the integral carry recurrence in the weighted binary setting [13]. For adjacent work on the binary expansion of the Erdős–Borwein constant, see Crandall and the subsequent result of Campbell [14, 15]; neither result is used or formalised in this development. We make no priority claim for these mathematical ingredients; the contribution here is their verified integration, the stated certificate reductions and bounds, and the formalisation lessons described below.

The Mersenne–Lambert ladder. $L(f) = \sum_{n\geq 1} \frac{f(n)}{2^n - 1}$

$L(\mu) = \frac{1}{2}$	$L(\varphi) = 2$	$L(\mathbf{1}) = E$	$L(\varphi * \mu) = S$ (#249)
rational	rational	**irrational (proved)**	**open**

Logical status of the #249 reduction.

$S \in \mathbb{Q} \implies$ eventual binary periodicity $\implies R_{N+h} - R_N \in \mathbb{Z}$ for the eventual period.

$$\mathsf{certifiedKill}(h, N, L) \implies R_{N+h} - R_N \notin \mathbb{Z}, \text{ and}$$
$$\exists L\, \mathsf{certifiedKill}(h, N, L) \iff R_{N+h} - R_N \notin \mathbb{Z}.$$

Solid implications and the equivalence are proved; the unresolved step is an unbounded supply of such non-integral tail differences.

2 The Mersenne–Lambert ladder

Define the Lambert-type transform of an arithmetic function f by

$$L(f) \;=\; \sum_{n\geq 1} \frac{f(n)}{2^n - 1} \;=\; \sum_{m\geq 1} \frac{(f * \mathbf{1})(m)}{2^m}, \qquad (f * \mathbf{1})(m) = \sum_{d|m} f(d),$$

the second equality being the standard Lambert rearrangement (one Dirichlet convolution by the constant $\mathbf{1} = \zeta$); Apostol supplies the classical arithmetic-function and Dirichlet-convolution background [6], and Merca–Schmidt give a modern treatment of Lambert-series factorisations [7]. Walking the ladder by convolution gives five values with rational, irrational, open, or transcendental status.

input f	value	mathematical status	treatment here
μ	$L(\mu) = 1/2$	rational	formalised
φ	$L(\varphi) = 2$	rational	formalised
$\mathbf{1}$	$L(\mathbf{1}) = E$	irrational	formalised
$A = \varphi * \mu$	$L(A) = S$	open (#249)	identities and bounds formalised
Id	$L(\mathrm{Id}) = \sum \sigma(m)/2^m$	transcendental	cited [10]

The weight $A = \varphi * \mu$ is the primitive-conductor count: $A(p) = p - 2$ on primes, $A \geq 0$, and $A * \zeta = \varphi$. One convolution by ζ separates the open constant S from the rational 2, and one more separates rational from transcendental ($\varphi * \zeta = \mathrm{Id}$, $\mathrm{Id} * \zeta = \sigma$). So S is the rung one convolution from the machine-checked irrational E: the same series $\sum(\cdot)/(2^d - 1)$, with the constant weight 1 replaced by the nonnegative weight A.

Proposition 2.1 (the rational rungs)**.** *$\sum_{d\geq 1} \mu(d)/(2^d - 1) = 1/2$ and $\sum_{d\geq 1} \varphi(d)/(2^d - 1) = 2$, exactly.*

Formalised: Lean and Lean; restated on the #249 series shape at Lean and Lean.

Proposition 2.2 (the positive lift, $L(A) = S$)**.** *With the primitive-conductor weight $A = \varphi * \mu$,*

$$\sum_{d\geq 1} \frac{A(d)}{2^d - 1} \;=\; \sum_{n\geq 1} \frac{\varphi(n)}{2^n} \;=\; S.$$

Formalised: Lean, with the weight A defined at Lean. This places S in the same $\sum(\cdot)/(2^d - 1)$ family as E, with a nonnegative weight.

Proposition 2.3 (the Möbius-square lens)**.**

$$S \;=\; \tfrac{1}{2} \;+\; \sum_{d\geq 1} \frac{\mu(d)}{(2^d - 1)^2}.$$

Formalised: Lean. Writing $T = \sum_{d\geq 1} \mu(d)/(2^d - 1)^2$, irrationality of S is equivalent to irrationality of T, since $S = \frac{1}{2} + T$. The factor $1/(2^d - 1)^2$ is the probability that d divides two independent fair-coin waiting times (Section 4.2). The signed constant, the coprime-pair mass, and the tail differences of Section 4.3 are three readings of the same quantity.

3 Erdős–Borwein-type irrationality (the #257 direction)

3.1 Full support: the Erdős–Borwein constant, every base

Theorem 3.1 (full-support irrationality)**.** *For every integer $b \geq 2$, the series $\displaystyle\sum_{n\geq 1} \frac{1}{b^n - 1}$ is irrational. In particular the Erdős–Borwein constant $E = \sum_{n\geq 1} 1/(2^n - 1)$ is irrational.*

Formalised: Lean (all bases) and Lean (the base-2 corollary). This is Erdős's 1948 theorem [1]. In the ladder of Section 2 it is the rung $L(\mathbf{1})$, and it is the fixed machine-checked point one convolution away from the open S.

The proof runs through the certificate machine of the kernel rather than through a direct digit argument: a bounded Bertrand/CRT frame on a first block, a middle-window average over divisor pairs with a pigeonhole selection, and an explicit closure of the parameters. The Lambert bridge $\sum 1/(b^n - 1) = \sum_m \tau(m)/b^m$ (divisor-count series) is Lean in the source.

3.2 Named infinite-support cases

Beyond full support, the development proves irrationality for a family of infinite supports A, at every base $b \geq 2$. Each is a formalised case of the #257 statement family. None is the universal statement, and none is claimed as new mathematics.

Theorem 3.2 (support-class family)**.** *For every integer $b \geq 2$, the series $\sum_{n\in A} 1/(b^n - 1)$ is irrational for each of the following supports A:*

(a) *factorial support $A = \{n! : n \geq 1\}$, giving $\sum 1/(b^{n!} - 1)$;*

(b) *power-of-two support $A = \{2^n : n \geq 0\}$, giving $\sum 1/(b^{2^n} - 1)$;*

(c) *multiples $A = d\mathbb{N}$;*

(d) *pairwise-coprime supports with summable reciprocals (Erdős's condition [2]);*

(e) *eventually-periodic supports, residue classes, and the odd numbers.*

These sort into three groups. The classical case is (d): infinite pairwise-coprime A with $\sum_{a\in A} 1/a < \infty$, formalised from Erdős's condition. The lcm-gap named cases are (a) and (b), whose support gaps outgrow the running lcm; the structured families are (c) and (e).[1] Formalised: (a) Lean; (b) Lean; (c) Lean; (d) Lean; (e) Lean, Lean, Lean. The two base-2 instances that name the problem directly are Lean and Lean.

The prime support is now known unconditionally: Tao and Teräväinen proved that

$$\sum_{p} \frac{1}{2^p - 1} = \sum_{n\geq 1} \frac{\omega(n)}{2^n}$$

is irrational [16]. Their proof uses quantitative two-point correlations of multiplicative functions and is not formalised here. This settles an important special case, while the universal support statement remains open.

The factorial and power-of-two cases are worth a word because they are not reachable by a Liouville shortcut: the power-of-two gap $2^k - \mathrm{lcm}$ stays within a bounded ratio of the lcm, so the terms do not decay fast enough for a transcendence-measure argument, and a uniform criterion is needed. The mechanism is *period noncollapse*: a prime-valuation-deficit witness forces the multiplicative order of the base modulo the relevant modulus not to collapse, which rules out the long repeated digit blocks a rational value would require.

3.3 What stays open

The universal Erdős #257 statement, irrationality of $\sum_{n\in A} 1/(2^n - 1)$ for *every* infinite A, is not proved. The proved cases are a family, not the quantified theorem.

4 The totient constant S (Erdős #249)

Whether $S = \sum_{n\geq 1} \varphi(n)/2^n$ is irrational is open. The development gives two things and claims neither as a solution: an unconditional exclusion of small-denominator rationals, and a conditional reduction to finite certificates.

4.1 Unconditional: no small denominator

Theorem 4.1 (denominator exclusion)**.** *Set*

$$Q_0 := 79\,639\,646\,646\,701\,375\,323\,355\,774\,875\,831\,053 \approx 7.96 \times 10^{34}.$$

For every integer p and every q with $1 \leq q \leq Q_0$,

$$S \neq \frac{p}{q}.$$

Equivalently, if S is rational then its reduced denominator exceeds that bound.

[1] Periodic-coefficient Lambert series have their own literature; see e.g. Luca–Tachiya [11]. The cases here are formalised in the repository's own framework and are not claimed as new results.

Formalised: Lean. The bound is reached by a ladder of finite, elementary exclusions,

$$4838 \;\to\; 4\,194\,304 = 2^{22} \;\to\; 2.49\times 10^{17}\ (\text{Farey } K{=}120) \;\to\; 7.96\times 10^{34}\ (\text{Farey } K{=}240),$$

each a mediant/Farey argument that traps any candidate m/q inside a forbidden gap. The mediant framework is historically associated with Farey's 1816 note on vulgar fractions [9]; the finite exclusions themselves are the Lean-checked arguments of this release. See Lean for the first rung and Lean, Lean for the two Farey windows.

This is an explicit lower bound on the denominator of any rational representation, not a proof of irrationality. A rational with a larger denominator is not excluded by it.

Remark (one bound, four representations). Because the ladder identities of Section 2 are equalities, the same finite record transfers verbatim to three other representations of the constant: to the positive lift $\sum_d A(d)/(2^d-1)$ (Lean), to the signed Möbius-square constant $T=\sum_d \mu(d)/(2^d-1)^2$ at half the bound (Lean), and to the coprimality probability of Section 4.2 at half the bound. Half appears because $S=\frac{1}{2}+T$ turns a denominator d for T into $2d$ for S.

4.2 The geometric picture: S as coprime-pair mass

Let X,Y be independent fair-coin waiting times, $\Pr(X=n)=2^{-n}$ for $n\ge 1$. Then $\Pr(d\mid X)=1/(2^d-1)$, so the Lambert transform is a divisor calculus of X: $L(f)=\mathbb{E}[(f*\zeta)(X)]$. The rungs read as $L(\mu)=\Pr(X=1)=1/2$, $L(\varphi)=\mathbb{E}[X]=2$, $L(\mathbf{1})=\mathbb{E}[\tau(X)]=E$, and $L(A)=\mathbb{E}[\varphi(X)]=S$.

Counting visible lattice points (a,b) with $a+b=n$, $a>0$, $\gcd(a,b)=1$ gives exactly $\varphi(n)$, uniformly in n, so S is itself a coprime-pair mass.

Proposition 4.2 (coprimality probability)**.**

$$S \;=\; \tfrac{1}{2} \;+\; \Pr\big(\gcd(X,Y)=1\big) \;=\; \tfrac{1}{2} \;+\; \sum_{\substack{a,b\ge 1\\ \gcd(a,b)=1}} 2^{-(a+b)}.$$

Formalised: Lean, with the underlying lattice identity at Lean. Base 2 is the one point where $\Pr(X=a)\Pr(Y=b)=2^{-(a+b)}$ needs no normalising constant, so the totient sum and the coprimality probability line up with no boundary correction. In this reading, Erdős #249 asks whether two independent fair-coin waiting times are coprime with irrational probability.

4.3 Conditional: reduction to finite certificates

The reduction uses only the elementary bound $\varphi(n)\le n$, geometric tail estimates, and finite integer arithmetic. For $t\ge 1$, write $M_t=\operatorname{lcm}(1,\dots,t)$.

Step 1: rationality forces an integer tail-period law. Write the fractional layer of $2^N S$ as the tail

$$R_N \;=\; \sum_{m\ge 1}\frac{\varphi(N+m)}{2^m}.$$

If S is rational, its binary expansion is eventually periodic, which forces, for a suitable period h and all large N, the integrality $R_{N+h}-R_N\in\mathbb{Z}$. To refute integrality at a finite depth L, form the window discrepancy

$$\Delta_{h,N,L}=\sum_{j=0}^{L-1}\big(\varphi(N+h+1+j)-\varphi(N+1+j)\big)\,2^{\,L-1-j}\in\mathbb{Z},$$

the first L binary digits of $R_{N+h} - R_N$ up to a controlled deep tail (bounded using only $\varphi(n) \le n$), and call the pair *certified* when the residue of $\Delta_{h,N,L}$ modulo 2^L misses a neighbourhood of 0:

$$\mathsf{certifiedKill}(h, N, L): \quad N + h + L + 2 \;<\; \left(\Delta_{h,N,L} \bmod 2^L\right) \;<\; 2^L - (N + h + L + 2).$$

This is a finite integer computation, decidable by the kernel. It is the natural finite object here because a rational S would pin the tail difference to an integer, and a residue that stays a fixed distance from 0 is a finite proof that it did not.

Corollary 4.3 (tail-period reduction)**.** *If for every period $h \ge 1$ and every threshold N_0 there exist $N \ge N_0$ and L with $\mathsf{certifiedKill}(h, N, L)$, then S is irrational.*

Formalised: Lean; the tail R_N and the predicate are Lean and Lean.

Step 2: collapse the two parameters to one. Every multiple of a period is a period, so $\operatorname{lcm}(1, \dots, t)$ is the universal-period envelope: any eventual period of a rational S divides it once t is large. Standing on that one diagonal ray covers all possible periods at once, and the position parameter drops out too.

Corollary 4.4 (diagonal reduction)**.** *Suppose that for every threshold t_0 there are parameters $t \ge t_0$ and L for which $\mathsf{certifiedKill}(M_t, M_t, L)$ holds. Then S is irrational.*

Formalised: Lean.

Step 3: the cone, flatness, and completeness. On the cone $\{kM_t : k \ge 1\}$, rationality forces not one pairwise relation but a single fractional constant across the whole cone.

Proposition 4.5 (cone flatness)**.** *If S is rational, then for every sufficiently large t there is $\theta_t \in [0, 1)$ such that the fractional part of the tail R_{kM_t} equals θ_t for every $k \ge 1$.*

Formalised: Lean.

The certificates are exact, not a lossy heuristic: a certificate exists exactly when the object it certifies is non-integral. Thus the finite computations are machine-checked witnesses rather than numerical experiments.

Proposition 4.6 (certificate completeness)**.** *For all h, N,*

$$\big(\exists L,\ \mathsf{certifiedKill}(h, N, L)\big) \quad\Longleftrightarrow\quad R_{N+h} - R_N \notin \mathbb{Z}.$$

Formalised: Lean. Finite certificates witness non-integrality. A joint refuter can interrogate a menu of cone vertices at once and is sharper than any pairwise test: it refutes the single-fractional-part model that rationality predicts on a finite sample of cone vertices (Lean).

The certificates are non-empty. The initial segment of the required supply is machine-checked, so the reduction is not vacuous. By completeness, each of these is a kernel-checked proof of a specific non-integral tail difference, not a numerical experiment.

Example 4.7 (verified finite instances)**.** The following are checked by the Lean kernel:

- $\mathsf{certifiedKill}(h, 12, 16)$ for every period $h \in [1, 8]$;
- certified kills for every period $h \in [1, 16]$ at a fixed window;
- diagonal certificates $\mathsf{certifiedKill}(M_t, M_t, L)$ for $t \in [1, 6]$, and at $t = 7, 8$;
- a tabulated joint cone-vertex certificate.

Formalised: Lean, Lean, Lean (with Lean, Lean), and Lean. These instances show that each layer of the reduction is non-vacuous and, by completeness, exact. They are not evidence that the unbounded supply exists.

What is open. The reduction converts #249 into the statement that the certificate supply is *unbounded*: certificates exist for arbitrarily large parameters. By Proposition 4.6 this is equivalent to the tail differences being non-integral infinitely often, and equivalently to the Farey gap denominators of Section 4.1 being unbounded: the same obstruction seen as a rational denominator, an eventual binary period, and a flat tail on the lcm cone. That unboundedness is the remaining analytic question, and it is left open. The reduction is real; the closure is not asserted.

5 Formalisation architecture and mathematical lessons

The formal development repeatedly separates a finite, kernel-checkable core from the analytic statement that would have to hold uniformly. For the full-support Erdős–Borwein theorem, the core is a bounded certificate engine assembled from a Bertrand/CRT frame, divisor-pair averaging, and an explicit parameter closure. For #249, the same design principle appears more sharply: eventual binary periodicity predicts an integral tail difference; finite modular arithmetic can refute that prediction at any fixed pair of parameters; and completeness proves that this computation loses no information. Lean therefore makes the open boundary visible: the missing ingredient is not another finite verification but a theorem producing witnesses at unbounded parameters.

The representation of eventual periodicity was itself consequential. Rather than reason about an informal repeating digit string, the development uses the scaled tails R_N and proves that rationality forces their period differences to be integers. Certificate completeness then separates two logically distinct tasks: finite evaluation decides non-integrality at fixed parameters, while an analytic theorem would still be needed to produce such parameters without bound. This separation is what prevents verified computation from being mistaken for the open theorem.

Three changes of coordinates serve different mathematical purposes. The Mersenne–Lambert transform places S next to the Erdős–Borwein constant and exposes the nonnegative primitive-conductor weight. The Möbius-square identity turns the same number into a signed rapidly convergent series. The coprime-pair identity turns it into a probability. Formalising all three and proving that the denominator exclusion transfers between them prevents an informal change of representation from silently changing the claim.

The auxiliary carry modules ask what rationality would force rather than claiming a new irrationality theorem. They give an exact tempered-orbit criterion, Boolean–Möbius coordinates for support series, constraints from residue wraps and reciprocal mass, and a greedy description of the value set. The sublogarithmic coverage theorem adds a further lesson: the elementary fixed-power estimate $\tau(n)^k \le (k^{2^k})^k n$ becomes useful only after it is transported through the binary tail and composed with the exact carry recurrence. Rationality then forbids long zero windows in f_A, uniformly in the support. In the reciprocal-mass argument, divergent series had to be kept as an explicit alternative rather than passed through the convention that a divergent `tsum` evaluates to zero. Together they rule out several tempting but false proof strategies—for example, rationality need not give a bounded carry alphabet—and identify the kind of infinite Boolean carry path that a future proof would have to exclude. The precise statements and source links are retained in Appendix A; their supporting role is why they are not a second main narrative here.

This architecture is reusable beyond the two named problems: choose a representation in which rationality imposes a rigid tail law, isolate a finite predicate that soundly refutes that law, prove the predicate complete at fixed parameters, and state the remaining uniformity theorem separately. The last step matters as much as the first three. It prevents a large verified finite range from being mistaken for the infinite theorem.

6 Artefact availability and verification

The Lean 4 proof assistant [17] and Mathlib library [18] check the development. The source, exact informal-to-formal links, and build metadata are available in the public repository `plectis-lean-erdos249-257`. This paper is pinned to the tagged release `v0.4.0`; each blue Lean link targets an exact declaration at that tag (for example, the base-2 corollary in Theorem 3.1 resolves to Lean). The package uses `leanprover/lean4:v4.29.1` and the Mathlib revision pinned by `lake-manifest.json`. The exact public artefact cited here is release `v0.4.0` [20].

There is no `sorry` or `admit` in the package. Finite computations use `decide`, whose proof terms are checked by the Lean kernel; the package does not use `native_decide`. Reproduction requires only the pinned toolchain and the ordinary project build:

```
lake exe cache get
lake build
```

The second command elaborates every declaration and kernel-checks every proof term against the pinned Mathlib dependency. Continuous integration executes the same build. Appendix B gives a compact map of the principal informal statements to declarations. The repository's claim registry (`docs/claims.json`) records each public claim once, and `scripts/check_release.py` verifies that this paper's source links, the repository documentation, and the release metadata all agree with it, including that every Lean link above resolves to the named declaration at the stated line of the released source.

7 Conclusion

The formalisation fixes two mathematical landmarks and two exact frontiers. It verifies the all-integer-base Erdős–Borwein theorem and several structured support cases toward #257. For #249 it proves that a rational value of S would have denominator exceeding Q_0, and it reduces irrationality to an unbounded supply of finite certificates whose fixed-parameter semantics are complete. The unresolved #249 step is precisely that unboundedness; the unresolved #257 step is the passage from the formalised support classes to every infinite support. Neither frontier is softened by the size of the verified artefact.

On the #257 side, rationality would impose two complementary forms of rigidity: divisor-coverage zero windows have length $o(\log N)$, whereas an infinite support forces the associated positive carry states to be unbounded. These statements constrain different objects and do not yet exclude a rational infinite-support value, but together they narrow the structure that any such value would have to exhibit.

The main lesson from formalisation is the alignment of representations and proof obligations. The Lambert, Möbius, probabilistic, periodic-tail, and carry descriptions are not competing stories: each makes a different rationality obstruction explicit. A future advance must exploit one of those obstructions uniformly, not merely extend the finite range already checked.

Statements and declarations

Artefact and data availability. The complete formal artefact is the tagged public release `v0.4.0` of the repository, including the Lean sources, the pinned toolchain and Mathlib manifest, the generated certificate files, and this manuscript's source. Section 6 gives the verification route.

Use of AI assistance. Large-language-model assistants were used during development for proof drafting, refactoring, and prose editing. This does not affect the trust story: every mathematical claim rests on the published Lean source, in which each proof term is checked by the Lean 4 kernel against the pinned Mathlib, with no `sorry`, `admit`, additional axioms, or `native_decide`.

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A Auxiliary rationality criteria from binary carry systems

Five further modules provide additional criteria. They prove no new irrationality. Instead they pin down, in machine-checked form, what a rational value of the series in question would have to look like: as an integer carry orbit, as a Möbius-inverted support coordinate, as a residue-wrap cycle modulo the odd part of the denominator, and as a point of a greedy subset-sum geometry. Everything in this section is an equivalence or an unconditional constraint on the rational side; nothing excludes a rational value of either open series, and each module states that boundary in its header. The prior-art boundary is theorem-family-specific. The integral carry recurrence has a direct antecedent in Wang–Grau Ribas [13]; the strict-tail geometry is recorded by Kovač–Tao [12]; Möbius inversion, repetend algebra, and divisor averaging are classical (see, for example, Apostol [6]). We make no claim of mathematical priority for the converse/rigidity, certificate-normal-form, or coupled reciprocal-mass/collision statements collected here.

A.1 Tail-orbit rigidity: a rationality criterion

For a coefficient sequence $c : \mathbb{N} \to \mathbb{N}$ with $c(n) \le n$, write

$$X_c = \sum_{n \ge 1} \frac{c(n)}{2^n}, \qquad T_c(N) = \sum_{j \ge 1} \frac{c(N+j)}{2^j},$$

the series and its scaled tail after N binary digits. Call an integer sequence u a *tempered orbit* for c with multiplier $v \ge 1$ if

$$u(N+1) = 2\,u(N) - v\,c(N+1) \text{ for all } N, \qquad \text{and} \qquad u(N)/2^N \to 0.$$

The recurrence is exact binary long division against the coefficient stream; the boundary condition forbids the homogeneous solution. Wang and Grau Ribas use the integral carry recurrence forced by rationality for the weighted binary special case $c(n) = nd_n$ [13]. The module abstracts that forward mechanism to arbitrary nonnegative $c(n) \le n$; its explicit converse and subexponential-orbit rigidity are the local additions, with no priority claim attached.

Theorem A.1 (tail-orbit rigidity)**.** *Let $c(n) \le n$ for all n. Every tempered orbit is the scaled tail, $u(N) = v\,T_c(N)$ for all N; and X_c is rational if and only if a tempered orbit exists for some multiplier $v \ge 1$.*

Formalised: `Lean` (rigidity) and `Lean` (the criterion), restated in Mathlib's irrationality vocabulary at `Lean`. The engine is one observation: a real orbit with $d(N+1) = 2\,d(N)$ and $d(N) = o(2^N)$ vanishes identically (`Lean`).

The tempered boundary is essential: the homogeneous parasite $N \mapsto 2^N$ can be added to any orbit without disturbing the recurrence, so positivity of an orbit certifies nothing by itself, and positivity is nowhere advertised as an equivalent criterion. The linear-growth hypothesis covers both constants introduced in Section 1: $\varphi(n) \le n$ for the #249 coefficients, and $f_A(n) \le \tau(n) \le n$ for the #257 support coefficients $f_A(n) = \#\{a \in A : a \mid n\}$. The modules below instantiate the criterion on supports; the totient instantiation is not taken here.

A.2 Boolean–Möbius carry coordinates (the #257 direction)

At base 2 the support series of Section 3 is itself a binary coefficient series, $\sum_{n\in A} 1/(2^n-1) = X_{f_A}$ (Lean). Two exact coordinate changes then turn a hypothetical rational value into a support-free object. The divisor transform and Möbius inversion below are classical Lambert-series infrastructure [6, 7]; the contribution here is their Lean-checked composition with the tempered carry interface.

Proposition A.2 (Boolean Möbius inversion, both directions)**.** *On positive integers, $f_A = \mathbf{1}_A * \zeta$ and $\mu * f_A = \mathbf{1}_A$, where $\mathbf{1}_A$ is the indicator of A. Conversely, every integer-valued arithmetic function f whose Möbius transform is Boolean, $(\mu * f)(n) \in \{0,1\}$ for all $n \geq 1$, satisfies $f = f_B$ for the support $B = \{n \geq 1 : (\mu * f)(n) = 1\}$ selected by that transform.*

Formalised: Lean, Lean, and, with no search or finiteness hypothesis, the converse Lean; the selected support is Lean.

On the carry side, Theorem A.1 applies at $c = f_A$: the support series is rational exactly when a tempered carry orbit for f_A exists (Lean); a displayed fraction p/q corresponds to an orbit U with multiplier q and initial value $U(0) = p$ (Lean); and exact division recovers the coefficient from the orbit, $\big(2U(N) - U(N+1)\big)/q = f_A(N+1)$ (Lean). The elementary divisor-pair bound $\tau(n) \leq 2\lfloor\sqrt{n}\rfloor$ (Lean) confines the tails to a square-root strip, $T_{f_A}(N) \leq 2\sqrt{N} + 4$ (Lean), hence every orbit state to $0 < U(N) \leq q\,(2\sqrt{N}+4)$ (Lean, Lean); and the strip is strong enough to re-derive the tempered boundary, so the analytic side condition can be traded for one inequality. The two coordinate changes compose.

Theorem A.3 (certificate-level equivalence)**.** *Fix an integer p and an integer $q \geq 1$. There is a nonempty support A of positive integers with $\sum_{n\in A} 1/(2^n-1) = p/q$ if and only if there is an integer sequence U with $U(0) = p$ such that: every $U(N)$ is positive; $U(N) \leq q\,(2\sqrt{N}+4)$; q divides $2U(N) - U(N+1)$ for every N; and the Möbius transform of the quotient sequence $\big(2U(N) - U(N+1)\big)/q$ is Boolean. The support is not guessed: it is reconstructed from the certificate by Proposition A.2.*

Formalised: Lean, with the certificate packaged as Lean and the quotient normalized at index zero (Lean).

Example A.4 (the support $\{2,3\}$)**.** The support $\{2,3\}$ has value $\frac{1}{3} + \frac{1}{7} = \frac{10}{21}$ (Lean), its orbit is the pure period-six cycle $10, 20, 19, 17, 13, 26$ with multiplier 21 (Lean), and the Möbius transform of the carry quotient recovers exactly $\{2,3\}$ (Lean).

These are coordinates, not by themselves a strategy: nothing here excludes infinite Boolean carry paths, and no finite search is promoted to a completeness argument.

A.3 Sublogarithmic divisor coverage

The coefficient $f_A(n)$ counts the elements of A dividing n. Say that a zero window of length h starts after $c + N$ when $f_A(c + N + j + 1) = 0$ for every $0 \leq j < h$. Thus no integer in that interval is divisible by an element of the support.

Theorem A.5 (sublogarithmic zero windows)**.** *Let A contain a positive integer and suppose*

$$\sum_{a\in A} \frac{1}{2^a - 1} = \frac{p}{2^c v}, \qquad p \in \mathbb{Z}, \quad c \in \mathbb{N}, \quad v \geq 1.$$

For every $\varepsilon > 0$ there is a constant $B \geq 0$, depending only on ε, c, and v, such that every zero window after $c + N$ satisfies

$$h \leq \varepsilon \log_2(N+1) + B, \qquad \log_2 x := \frac{\log x}{\log 2}.$$

The bound is uniform in A, p, N, and h.

Formalised: Lean.

The proof first establishes, for each integer $k \geq 2$, the explicit divisor estimate

$$\tau(n)^k \leq \left(k^{2^k}\right)^k n.$$

It transports the resulting $n^{1/k}$ bound through the binary tail and then uses the exact carry recurrence across a zero window. Raising the resulting inequality to the kth power absorbs the occurrence of h on the right and leaves a coefficient $1/(k-1)$ in front of the binary logarithm. Choosing k after ε gives the theorem. This controls gaps in divisor coverage; it does not assert a subpower bound for the carry state itself.

A.4 Residue wraps, reciprocal mass, and unbounded carries

Suppose now the support series equals $p/(2^c v)$ with v odd and A containing a positive element. Rationality makes the scaled tails integers: $u(N) = v\, T_{f_A}(c+N)$ is a positive integer, satisfies the carry recurrence $u(N+1) + v\, f_A(c+N+1) = 2\, u(N)$, and reduces modulo v to the doubling residues $r_N = p\, 2^N \bmod v$ (Lean). Doubling a least residue either stays under v or wraps past it exactly once, so each step emits a wrap digit $k_N = \lfloor 2r_N/v \rfloor \in \{0,1\}$, and over any cycle length h with $2^h \equiv 1 \pmod{v}$ the repetend identity holds:

$$\sum_{N<h} r_N \;=\; v \cdot w, \qquad w = \sum_{N<h} k_N,$$

with $w \geq 1$ when p is coprime to $v > 1$. Formalised: Lean and Lean; residues, digits, and counts are Lean, Lean, Lean.

Proposition A.6 (one-wrap classification)**.** *Let $v > 1$ be odd, p coprime to v, and $h \geq 1$ with $2^h \equiv 1 \pmod{v}$. If the cycle of p wraps exactly once, then $v = 2^h - 1$ and the starting residue is a power of two: $r_0 = 2^a$ for some $a < h$.*

The one-wrap cycles are thus exactly the Mersenne repetends $2^a/(2^h - 1)$. Formalised: Lean, from the generic closed-cycle form Lean, and instantiated at the true order $h = \operatorname{ord}_v(2)$ (Lean, with Lean). The classification is algebraic; a finite check of 446 coprime starting residues across twelve modulus/order rows is retained as kernel-checked validation, not as the proof (Lean, Lean, Lean).

The analytic bridge is Cesàro averaging. Write $\rho(A) = \sum_{a\in A} 1/a$ for the reciprocal mass (Lean). When the reciprocal family is summable, the mean of $f_A(1), \ldots, f_A(N)$ converges to $\rho(A)$ — the density of the multiples of a, summed over the support — and the mean of the scaled tails has the same limit (Lean, Lean). The divergent case is carried as an explicit disjunction, not through the convention that a divergent `tsum` is zero (Lean). Each tail state splits exactly as $u(N) = v\, e_N + r_N$ with integral excess $e_N = \lfloor u(N)/v \rfloor \geq 0$ (Lean).

Proposition A.7 (order-wrap lower bound, with exact excess mean)**.** *Let A and $p/(2^c v)$ be as above with $v > 1$ odd, and let w be the wrap count of p over one cycle of length $h = \operatorname{ord}_v(2)$. Then $\sum_{a\in A} 1/a$ diverges or $\rho(A) \geq w/h$; if moreover p is coprime to v, then $w \geq 1$, so $\rho(A) \geq 1/\operatorname{ord}_v(2)$. In the summable case the bound is the periodic part of an exact identity: the Cesàro mean of the excess e_N converges automatically, and*

$$\rho(A) \;=\; \frac{w}{h} \;+\; \lim_{N\to\infty} \frac{1}{N} \sum_{n<N} e_n.$$

Formalised: Lean, Lean, the form retaining the divergent alternative Lean, and the excess mean Lean (fraction-facing shifted form Lean, named-limit form Lean). A rational value thus ties the odd part of its denominator to the sparsity of its support: a summable support with small $\rho(A)$ can only display denominators whose odd part has small multiplicative order of 2.

Corollary A.8 (collision strengthening at common multiples)**.** *In the summable case, let $F \subseteq A$ be finite and let L be a common multiple of F. Then*

$$\rho(A) \;\geq\; \frac{w}{h} \;+\; \frac{\lceil (|F|-1)/2 \rceil}{L}.$$

In particular, an infinite support whose value is a dyadic rational $p/2^c$ has $\sum_{a\in A} 1/a$ divergent or $\rho(A) > 1$.

The mechanism is a collision: at every positive multiple of L at least $|F|$ support divisors land on the same index, so $f_A \geq |F|$ there, while the exact carry equation $f_A(n)+e_n = k_{n-1}+2e_{n-1}$ with $k_{n-1} \leq 1$ lets one step absorb at most one unit — so the preceding excess must spike to at least $\lceil (|F|-1)/2 \rceil$, and the spikes recur with density $1/L$. Formalised: Lean (denominator-shifted form Lean), the carry equation Lean, and the dyadic corollary Lean (via Lean).

Theorem A.9 (carry states are unbounded over infinite supports)**.** *Let A be infinite with value $p/(2^c v)$, $v \geq 1$. Then the positive integer carry state $u(N) = v\, T_{f_A}(c+N)$ is unbounded: for every B there is an N with $u(N) > B$.*

The proof picks $2B+1$ positive support elements; at a common multiple L of all of them, the local inequality $1 + v\,|F| \leq 2\,u(L-c-1)$ pushes the state past B. Formalised: Lean, from Lean and the local inequality Lean; the $\{2,3\}$ fixture revalidates it at sixty-six common multiples (Lean). So a rational value over an infinite support cannot run on a bounded carry alphabet. That closes finite-state readings of the carry system; it does not close the problem. Nothing in the constraints above prevents an infinite support from satisfying every one of these constraints. The repetend identities and divisor averages used here are classical. We make no claim of mathematical novelty for the coupled reciprocal-mass bounds, collision strengthening, or global unboundedness statement.

A.5 Greedy geometry of the #257 value set

The fourth module studies the set of all values a base-2 support series can take. For $n \geq 1$ put $w_n = 1/(2^n - 1)$, let $T(n)$ be the mass strictly after exponent n, and let

$$\mathcal{A} \;=\; \Big\{ \sum_{n\in A} w_n \;:\; A \subseteq \mathbb{N},\; 0 \notin A \Big\},$$

the *Mersenne achievement set*, the exponent 0 being normalised away as analytically invisible. Coding by supports of positive exponents is literally the formal support series (Lean; the set is Lean). In this language the base-2 #257 statement reads: the rational members of $\mathcal{A}$ are exactly the finite subset sums. Nothing below decides that. Kovač and Tao record the strict separation, uniqueness, and Cantor-set geometry for these Lambert subseries [12]; this is the relevant Lambert-series instance of Kakeya's classical strict-tail criterion for subsum sets [8]. The local contribution is the executable Lean interface, quantitative enclosure, and finite non-membership certificates over that known geometry.

Theorem A.10 (superincreasing geometry and greedy membership)**.** *For every $n \geq 1$ the weight dominates the whole tail after it, $T(n) < w_n$, with the two-scale gap $w_n - T(n) = \frac{2}{3}4^{-n} + O(8^{-n})$ and explicit valid O-constant 3. Consequently a real x belongs to $\mathcal{A}$ exactly when $x \geq 0$ and every greedy residual of x is at most the remaining tail; and normalised support coding is injective, so each member of $\mathcal{A}$ has a unique support, which the greedy recursion recovers bit by bit.*

Formalised: Lean, Lean (explicit bound Lean), Lean, and Lean, transferred to the kernel series at Lean. The nonnegativity guard in the membership criterion is necessary: without it a negative target would survive every level while lying outside $\mathcal{A}$.

Proposition A.11 (exact rational runs and finite non-membership certificates)**.** *The greedy recursion executed in exact rational arithmetic agrees step by step with the real recursion under casting. A finite decidable certificate — the exact rational residual at some level is at least an exact rational upper enclosure of the remaining tail — soundly proves non-membership in $\mathcal{A}$; the level-one instance shows $3/4 \notin \mathcal{A}$. And a rational number already known to lie in $\mathcal{A}$ has computable support bits.*

Formalised: Lean; Lean, with the certificate Lean and the enclosure Lean; Lean; and Lean. The certificates are one-sided: survival through any finite depth proves nothing about membership, and no decidability of membership in $\mathcal{A}$ is asserted. We do not formalise the topological or measure-theoretic geometry of $\mathcal{A}$; in particular, no "fat Cantor set" statement is claimed. These criteria do not exclude rational values for either open series.

B Index of principal declarations

Each row links to the declaration at release `v0.4.0`.

informal statement	linked Lean module
$\sum 1/(b^n - 1)$ irrational, all $b \geq 2$	Lean (`CertificateKernel`)
Erdős–Borwein E irrational	Lean (`CertificateKernel`)
factorial support	Lean (`CertificateKernel`)
power-of-two support	Lean (`CertificateKernel`)
pairwise-coprime supports	Lean (`CertificateKernel`)
$L(\mu) = 1/2$	Lean (`MersenneLambertLadder`)
$L(\varphi) = 2$	Lean (`MersenneLambertLadder`)
positive lift $L(A) = S$	Lean (`CertificateKernel`)
Möbius-square lens	Lean (`CertificateKernel`)
$S = \frac{1}{2} + \Pr(\gcd = 1)$	Lean (`CertificateKernel`)
$S \neq p/q$ for $q \leq Q_0$	Lean (`CertificateKernel`)
tail-period reduction	Lean (`TotientTailPeriodKiller`)
diagonal reduction	Lean (`LcmDiagonalReduction`)
cone flatness from rationality	Lean (`LcmConeFlatness`)
certificate completeness	Lean (`LcmConeFlatness`)
joint cone refuter	Lean (`LcmConeNonflat`)
verified finite instances	Lean (`TotientTailPeriodKiller`)
tail-orbit rationality criterion	Lean (`GenericTailOrbitRigidity`)
Boolean carry certificate $\Leftrightarrow$ support fraction	Lean (`BooleanMobiusCarry`)
one-wrap classification	Lean (`RationalSupportCarrySkeleton`)
order-wrap mass bound	Lean (`RationalSupportCarrySkeleton`)
unbounded carry states	Lean (`RationalSupportCarrySkeleton`)
sublogarithmic zero windows	Lean (`SublogDivisorCoverage`)
greedy membership criterion	Lean (`GreedyAchievementSet`)

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